

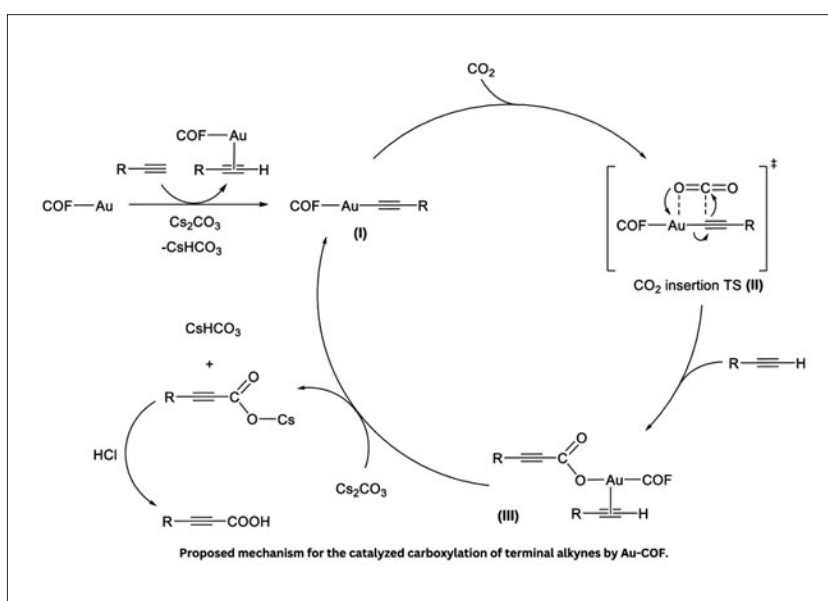
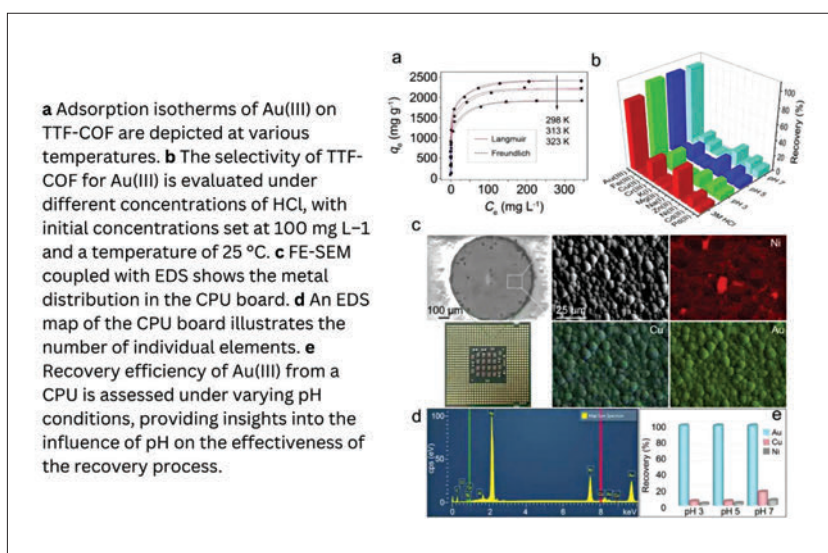
present. This selectivity is a big deal because it minimises the need for additional processing steps.

Moreover, the robustness of TTF-COF ensures it can be reused multiple times without losing efficiency. Its tightly packed crystalline structure maintains consistent performance, even under challenging conditions. On the other hand, TPE-COF, while still effective, doesn't quite match up due to its less sulfur-dense framework. But the innovation doesn't stop at gold recovery. Once loaded with gold, these COFs are repurposed as catalysts for the carboxylation of terminal alkynes – a fancy way of saying they help convert carbon dioxide into valuable organic compounds. This reaction is a cornerstone of green chemistry, tackling CO<sub>2</sub> emissions while producing useful materials.

Under modest conditions – ambient CO<sub>2</sub> pressure and moderate temperatures – gold-loaded COFs transform alkynes into propiolic acids with exceptional yields. What's more, these COFs retain their catalytic prowess over multiple cycles, proving their durability and cost-effectiveness. Unlike traditional catalytic systems relying on rare metals like platinum or silver, COFs offer a more sustainable and affordable alternative.

## Bridging waste management and clean energy

What makes this approach so revolutionary is its dual impact: reducing e-waste while advancing clean energy solutions. By reclaiming gold from discarded electronics, researchers address a massive environmental issue and create a platform for sustainable innovation. At the same time, these COFs align with global efforts to repurpose CO<sub>2</sub>, turning a greenhouse gas into a resource. The practical advantages are equally compelling. COFs are synthesised using relatively simple



methods, such as solvothermal reactions, and rely on readily available precursors. This scalability makes them attractive for industrial applications. Moreover, their stability and reusability mean they won't need frequent replacement, reducing long-term costs.

Despite their promise, COFs aren't without challenges. Scaling their production to industrial levels is no small feat. Current synthesis methods, while effective, require optimisation for speed and cost efficiency. Researchers are exploring innovations like automated pro-

duction and one-pot reactions to simplify the process.

Then there's the question of e-waste collection. Efficiently gathering and preprocessing discarded electronics varies widely by region, influenced by infrastructure and local policies.

Standardised recycling protocols and financial incentives could go a long way in addressing this issue. Additionally, integrating COF-based recovery systems into existing recycling setups will require cooperation between policymakers, scientists, and industry leaders. **[1]**